Research agent

Two agents search and screen the literature.
A person decides what is kept.



plan screen decide

Unit 11: Individual project presentation
Intelligent Agents, University of Essex Online
October 2026

The design from Unit 6

Goal	Planning Agent	Sources	Curation Agent	Reviewer	Result
Research question and criteria	Plans and revises searches	PubMed + Semantic Scholar; Crossref: status	Recommendation, reason and extract	Approves, flags or rejects	Approved papers stored

Checked JSON handover; human approval before storage.



Selected classes and fields: request, shared state and run budget

BDI-inspired state: Rao and Georgeff (1995). Goal-based agent: Russell and Norvig (2022).
Action and observation loop: Yao et al. (2023).

Design to code: kept, changed, why

Kept

Two agents and a checked JSON handover

Human approval before storage

PubMed and Semantic Scholar; Crossref for status

Gemini behind a replaceable adapter

SQLite, Streamlit and LangGraph

Changed	Why
Full text omitted	Unused source removed. Screening uses titles and abstracts.
18 model requests per run	Bounds each run. Daily account usage is not tracked.
Timeouts and counted retries	Busy or stalled requests led to bounded recovery.
Live run set to Flash-Lite	The eighth attempt completed. Flash remains the code default.



Changes and reasons recorded against the Unit 6 proposal.

The agent at work: a goal becomes a plan

Live goal, 18 September

Did the study evaluate ivermectin as a COVID-19 treatment and report clinical, virological or safety outcomes relevant to that treatment?

1 PubMed

ivermectin AND COVID-19 AND
(treatment OR clinical trial OR
randomized controlled trial OR outcome)

Finished: 50 papers

2 Semantic Scholar

ivermectin covid-19 treatment
clinical virological safety

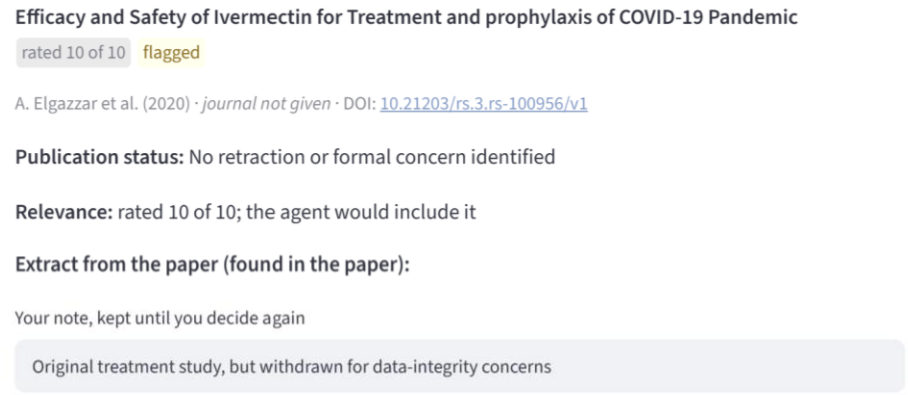
Finished: 50 papers

Goal to plan to retrieval. After a search: continue, refine, drop or finish, with a reason.
Every model request counts towards the default run budget of 18.

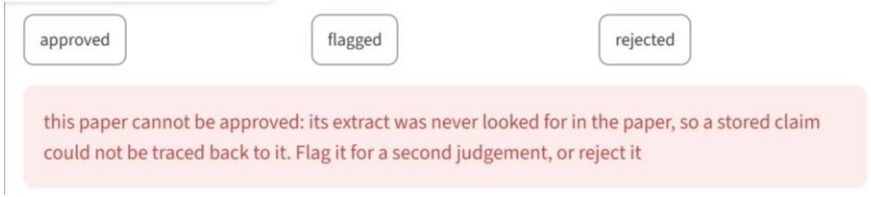


Live goal and two completed searches. Run 98af89fa5cdc, 18 September.
Text transcribed from the submitted run log; these entries are not next-move decisions.

Retrieval, screening and the extract check



Live review excerpts, 18 September: withdrawn preprint, relevance 10/10; I flagged it.
Elgazzar et al. (2020); withdrawal: Research Square (2021).



Invented paper: approval refused beside the decision

1 Retrieval

Matched by identifier, then title. 50 screening slots shared by turns across searches. Gaps stay visible.

2 Screening

Batches of ten. Include, exclude or uncertain, with a score, reason, summary and extract.

3 The check

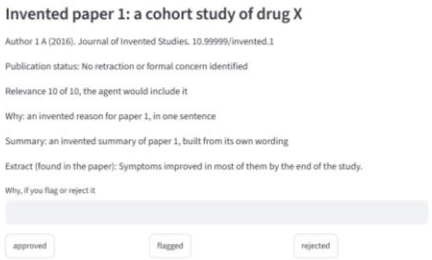
The extract must occur in the title or abstract. Approval is blocked otherwise.

4 Guardrails

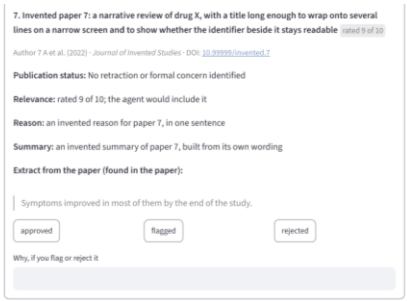
Paper text is marked as data. Prompt boundaries are tested; meaning still needs review.

Search reporting: Page et al. (2021); Rethlefsen et al. (2021)

The reviewer's gate, and what it cannot catch



Before, 16 September: extract and decision controls



After: a labelled card and separate extract

Observed

One participant: me.
Sessions on 16 and 19 September.

Changed and checked

Clearer cards and revision feedback.
Follow-up tasks: about 7–11 seconds.
Ease rating: 3 to 2; 1 is easiest.
Timings are not directly comparable.

Limit of the human gate

Lower-scoring papers can be hidden.
Approval controls what is stored;
it cannot recover papers never shown.

Evidence of execution

100
papers retrieved
50 from each search source

50
papers screened
50 places shared by turns

13
model requests
including revision

Live review

41 papers judged again; 4 recommendations changed.
Nine approvals matched nine stored identifiers.

CLEF: 50 records

Retrieval: 10/15, partial. Flash-Lite screening: precision 6/6; relevant found 6/15. Nine relevant papers were not recommended for inclusion.

CLEF-TAR (2019). Full methods and limits in the testing account.



Result: nine papers approved and saved

Development benchmark: 100 records

8/8 relevant retrieved at 1,000 requested results per search;
earlier run at 50: 0/8.
Known labels; separate runs.

Testing, and what it fixed



1,212

automated checks passed
3 skipped; 1 deselected
1 October 2026; Ruff clean

Why storage checks the extract again

```
# That check is made here rather than read off the record. `extract_found` says
# what was true when the run ended, and a judgement can be edited afterwards:

if not is_in_the_paper(item.screening.extract, item.paper):
    raise RefusedByStorage(
        f"the extract now on {_naming(item.paper)} is not in the paper, "
        "whatever the record of the earlier check says")
```

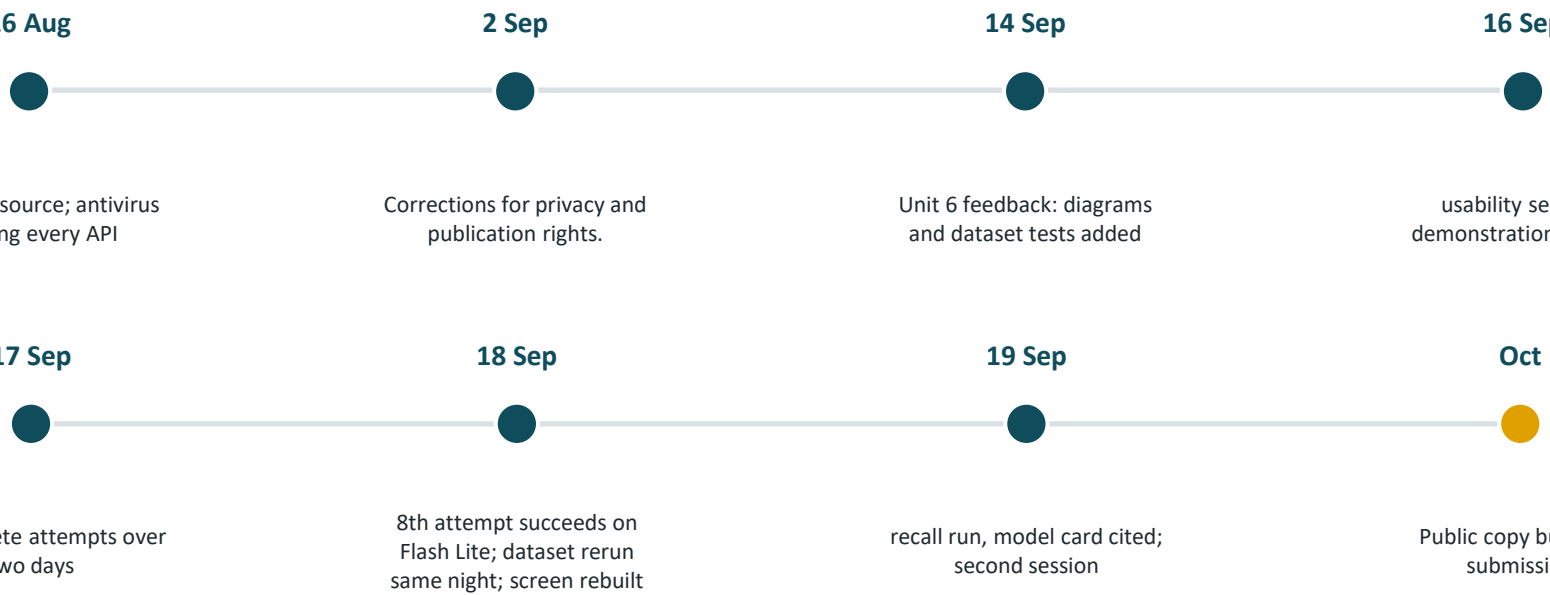
Comment and code excerpts, processing/storage.py

34 recorded remediation entries

R-21 Sensitive text escaped approval	Recursive rewriting and key checks	Six hostile approval tests
R-24 Invalid approval blocked the batch	Refusal beside the paper; storage guard kept	App and workflow tests
R-25 A model request kept waiting	Connection timeout and counted retries	Timeout and request-count tests

Unit, integration and functional checks use invented replies. Live audits are separate.
Methods: docs/testing.md. Decisions and remediation: docs/development.md.

How the work developed over the module



The development document records the decisions, problems and corrections.

What the system achieves, and its limits

Conclusion

A working chain from a research question to searches, screening and stored approvals.

One completed live workflow, small datasets and one familiar participant.

Relevant papers can be missed or hidden. Completeness is not established.

Next: a new topic with unseen labels, then testing with new reviewers.

Legal, social and ethical

UNESCO (2022) principles, mapped in the README.
Answered least well: fairness, on an uncalibrated ranking.

Key references

Rao and Georgeff (1995)

BDI agents: from theory to practice. ICMAS-95, pp. 312–319.

Yao et al. (2023)

ReAct: synergizing reasoning and acting in language models. ICLR 2023.

Page et al. (2021)

The PRISMA 2020 statement. International Journal of Surgery, 88, 105906.

Rethlefsen et al. (2021)

PRISMA-S. Journal of the Medical Library Association, 109(2), pp. 174–200.

Full references in the transcript.
Data: PubMed, Semantic Scholar, Crossref and CLEF TAR.
Software, models and AI acknowledgements: README.